Z-Scores & Linear Transformations

Section 1: Calculating and Interpreting Z-Scores

1 A student scores 82 on an exam where the class mean is 74 and the standard deviation is 5. What is the student's standardized z-score?

a. 1.60

b. 1.20

c. 0.80

d. -1.60

2 What does a z-score of -2.4 indicate about an observation?

a. The observation is 2.4 standard deviations below the mean

b. The observation is 2.4 units below the mean

c. The observation is in the 2.4th percentile

d. The probability of the observation is negative

3 On Test 1, Alice scored 88 (mean 80, SD 4). On Test 2, she scored 92 (mean 85, SD 5). On which test did she perform relatively better compared to peers?

a. Test 1 ($z_1 = 2.0 > z_2 = 1.4$)

b. Test 2 ($z_2 = 1.4 > z_1 = 1.0$)

c. Both performances were identical

d. Test 2 because her raw score was 92 vs 88

Section 2: Linear Transformations - Adding a Constant

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4 A dataset has mean $\mu = 50$ and standard deviation $\sigma = 8$. If 5 points are added to every observation ($y = x + 5$), what are the new mean and standard deviation?

a. New mean $= 55$, New SD $= 8$

b. New mean $= 55$, New SD $= 13$

c. New mean $= 50$, New SD $= 13$

d. New mean $= 55$, New SD $= 40$

5 If a constant $c = 10$ is subtracted from every value in a dataset, which statistic changes?

a. Median

b. Range

c. IQR

d. Variance

6 A teacher curves a test by adding 8 points to all student scores. How does this affect the shape of the score distribution?

a. The shape remains exactly the same

b. The distribution becomes more symmetric

c. The distribution becomes skewed to the right

d. The spread increases

Section 3: Linear Transformations - Multiplying by a Constant

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7 A dataset of temperatures in Celsius has mean 20°C and standard deviation 4°C. Converting to Fahrenheit using $F = 1.8C + 32$, what are the new mean and standard deviation?

a. Mean $= 68^\circ\text{F}$, SD $= 7.2^\circ\text{F}$

b. Mean $= 68^\circ\text{F}$, SD $= 39.2^\circ\text{F}$

c. Mean $= 36^\circ\text{F}$, SD $= 7.2^\circ\text{F}$

d. Mean $= 68^\circ\text{F}$, SD $= 4.0^\circ\text{F}$

8 If all observations in a dataset with variance 16 are multiplied by -3, what is the new variance?

a. 144

b. -48

c. 48

d. 12

9 If every observation is multiplied by 0.5, how does the IQR change?

a. The IQR is halved (multiplied by 0.5)

b. The IQR is unchanged

c. The IQR is quartered (multiplied by 0.25)

d. The IQR doubles

Section 4: Standardizing an Entire Distribution

10 If an entire non-normal distribution with mean $\mu = 40$ and SD $= 6$ is converted to z-scores ($z = \frac{x - 40}{6}$), what are the mean and standard deviation of the standardized values?

a. Mean $= 0$, Standard Deviation $= 1$

b. Mean $= 40$, Standard Deviation $= 1$

c. Mean $= 0$, Standard Deviation $= 6$

d. Mean $= 1$, Standard Deviation $= 0$

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11 Does standardizing a skewed dataset into z-scores transform the distribution into a normal shape?

a. No, standardizing changes center and scale but preserves the original skewed shape

b. Yes, standardizing always creates a standard normal bell curve

c. Only if the sample size is greater than 30

d. Yes, because the mean becomes 0 and standard deviation becomes 1

12 A runner's race time is at the 85th percentile of all competitors. What does this mean?

a. The runner was faster than 85% of competitors (finished with lower time than 15% of competitors)

b. The runner scored an 85% on the race

c. The runner finished in the top 85%

d. 85% of competitors finished before this runner

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